



Arrays and Strings

Managing Multiple Data Items in C Programming



Arrays



Strings



Initialization





Introduction



As programs become larger and more complicated, it becomes increasingly difficult to manage the data. Variable names typically become longer to ensure their uniqueness.

Solution: Arrays

Arrays provide a mechanism for declaring and accessing several data items with only one identifier, thereby simplifying task of data management.



Multiple Related Data

Many programs require processing of multiple, related data items like list of numbers, marks in a course, or enrollment numbers.



Alternative vs. Arrays

This could be done by creating several individual variables, but this is a hard and tedious process.





Array Declaration



Arrays are a kind of data structure that can store a fixed-size sequential collection of elements of the same type.

Concept

An array is used to store a collection of data, but it is often more useful to think of an array as a collection of variables of the same type.

◆ Individual Variables vs. Array

Instead of declaring individual variables like `number0`, `number1`, ..., `number99`, you declare one array variable like `numbers` and use `numbers[0]`, `numbers[1]`, ..., `numbers[99]`.

◆ Indexing

A specific element in an array is accessed by an index.





Syntax of Array Declaration



To declare an array in C, a programmer specifies the type of elements and the number of elements required by an array.

Syntax:

```
data_type arrayName [arraySize];
```

This is called a single-dimensional array.



Array Size

The arraySize must be an integer constant greater than zero.



Data Type

Type can be any valid C data type.

Example Declarations:

```
int char[80];  
float farr[500];
```



```
static int iarr[80];  
char chararray[40];
```

Size Specification

 The size of an array should be declared using a symbolic constant rather than a fixed integer quantity.

Advantages of Symbolic Constants:

- ✓ Makes it easier to modify a program that uses an array
- ✓ All reference to array size can be altered simply by changing the value of symbolic constant

```
#define SIZE 50 /* SIZE is a symbolic constant */
```





Example: This example shows how to declare and read values in an array to store marks of students of a class.



Accessing Array Elements



An element is accessed by indexing the array name. This is done by placing the index of the element within square brackets after the name of the array.

Example:

```
double salary = balance[9];
```

The above statement will take the 10th element from the array and assign its value to the salary variable.





Key Point: Array indices start from 0, so `balance[9]` is actually the 10th element.



Array Initialization



Arrays can be initialized at the time of declaration. The initial values must appear in the order in which they will be assigned to individual array elements, enclosed within braces and separated by commas.

Syntax:

```
data_type array-name[size] = {val1, val2, .....valn};
```

`val1` is the value for the first array element, `val2` is the value for the second element, and `valn` is the value for the `n`th array element.



Note: When initializing values at the time of declaration, there is no need to specify

the size.



Initialization Examples

```
int digits[10] = {1,2,3,4,5,6,7,8,9,10};  
int digits[] = {1,2,3,4,5,6,7,8,9,10};  
int vector[5] = {12,-2,33,21,13};  
float temperature[10] = {31.2, 22.3, 41.4, 33.2, 23.3, 32.3, 41.1, 10.8, 11.3, 42.3};  
double width[] = {17.33333456, -1.212121213, 222.191345};  
int height[10] = {60, 70, 68, 72, 68};
```



Key Point: Different data types can be initialized with appropriate values, and the size can be omitted when initialization is provided.



Character Arrays (Strings)



The array of characters is implemented as strings in C. Strings are handled differently as far as initialization is concerned.

Special Character:

A special character called null character '\0' implicitly suffixes every string.

◆ String Assignment

When external or static string character array is assigned a string constant, size specification is usually omitted and is automatically assigned.

◆ Null Termination

It will include '\0' character, added at end.

```
char thing[3] = "TIN";  
char thing[] = "TIN";
```





Key Difference: The first statement is not a string but simply an array storing three characters 'T', 'I' and 'N', whereas the second one is a four character string TIN\0.



Complete Example



Program to read values in an array and display them.

```
#include
#define SIZE 5 /* SIZE is a symbolic constant */

int main() {
    int i = 0; /* Loop variable */
    int stud_marks[SIZE]; /* array declaration */

    /* enter values of elements */
    for(i = 0; i < SIZE; i++) {
        printf("Element no. = %d\n", i + 1);
        printf("Enter value of element: ");
        scanf("%d", &stud_marks[i]);
    }

    printf("\nFollowing are values stored in corresponding array elements:\n\n");
    for(i = 0; i < SIZE; i++) {
        printf("Value stored in stud_marks[%d] is %d\n", i, stud_marks[i]);
    }

    return 0;
```

```
}
```

Output:

```
Element no. = 1 Enter value of element: 11
Element no. = 2 Enter value of element: 12
Element no. = 3 Enter value of element: 13
Element no. = 4 Enter value of element: 14
Element no. = 5 Enter value of element: 15
```

Following are values stored in corresponding array elements:

```
Value stored in a[0] is 11
Value stored in a[1] is 12
Value stored in a[2] is 13
Value stored in a[3] is 14
Value stored in a[4] is 15
```



Summary



Arrays

Store fixed-size sequential collection of elements of same type



Strings

Character arrays terminated with null character



Initialization

Can be done at declaration with braces



Arrays simplify data management by providing a mechanism for declaring and accessing several data items with just one identifier!

